GlucoTwin

A Digital Twin that warns of glucose excursions 2 hours ahead by fusing EHR and wearable / CGM data (Type 2 Diabetes)

Happiest Health · Reimagining and Reforming Healthcare in India Summit 2026 · Digital Twin Challenge

Team: <TEAM NAME> · College: <COLLEGE NAME>

Why diabetes needs a twin, not a snapshot

100M+

people with diabetes in India (ICMR-INDIAB estimate)

Minutes

of doctor time every few months, per patient

5 min

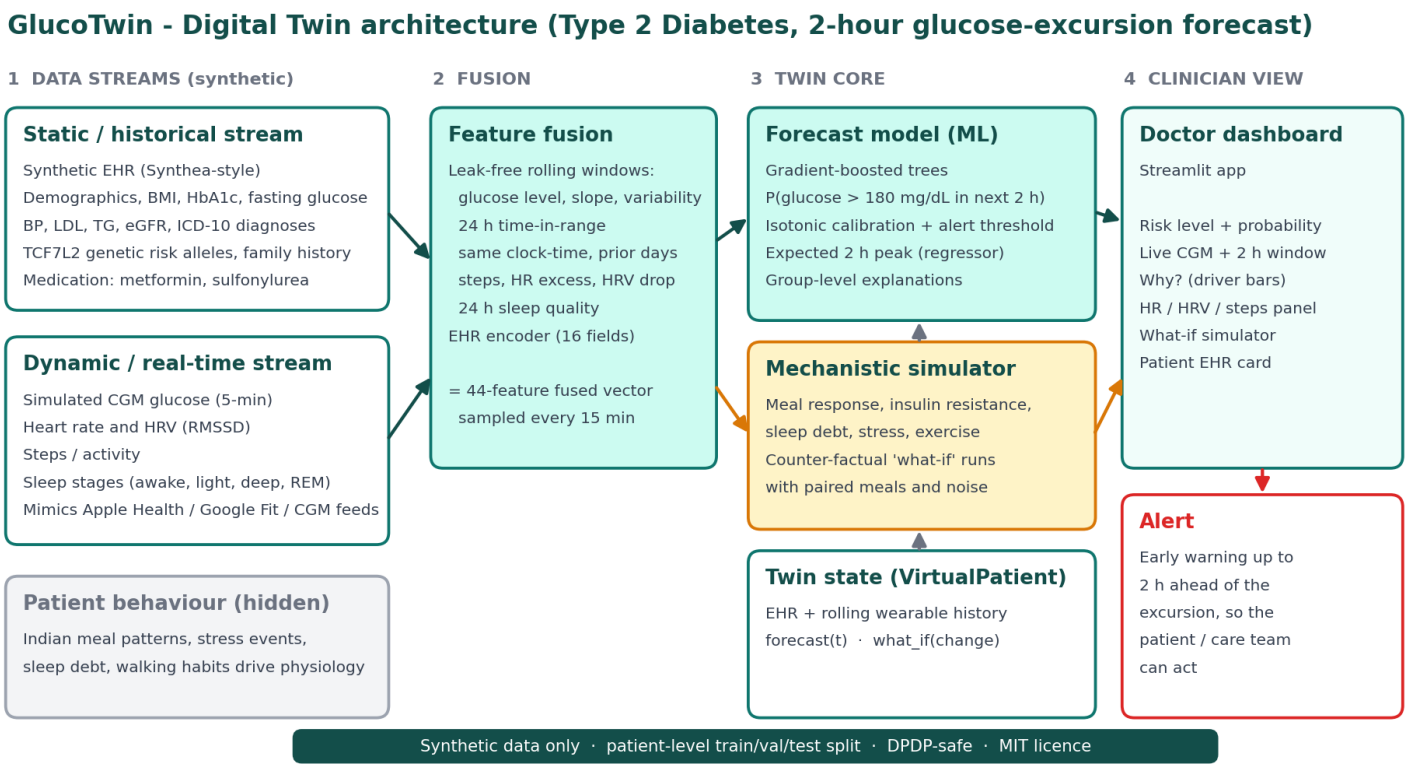
how often glucose, heart rate and sleep data change on a wearable

Today's care reacts after the spike. A Digital Twin keeps a living virtual replica of each patient, updated by wearables and joined with the health record, so the care team can act before an adverse event.

Use case: forecast, explain, test a fix

Condition	Type 2 Diabetes
Adverse event	Glucose above 180 mg/dL at any time in the next 2 hours (currently at or below 160)
User	Diabetologist or care team looking at a virtual patient console
Output	Probability + expected peak + risk level + 'why' + what-if simulation

Architecture



Two data streams, both synthetic

Static / historical (EHR)

- Age, sex, BMI, HbA1c, fasting glucose
- BP, LDL, triglycerides, eGFR
- ICD-10 diagnoses, medications
- TCF7L2 genetic risk alleles, family history

Dynamic / real-time (wearables)

- CGM glucose every 5 minutes
- Heart rate and HRV (RMSSD)
- Steps and activity
- Sleep stages (awake, light, deep, REM)

150 virtual patients × 14 days × 5-minute resolution. Indian meal patterns, sleep debt, stress and walking habits drive the physiology. No real patient data (DPDP / HIPAA safe).

The twin is a hybrid: data-driven + mechanistic

1

Forecast (ML)

Gradient-boosted trees on 44 fused, leak-free features. Calibrated probability, expected 2 h peak, alert threshold tuned for about 80% sensitivity.

2

What-if (simulator)

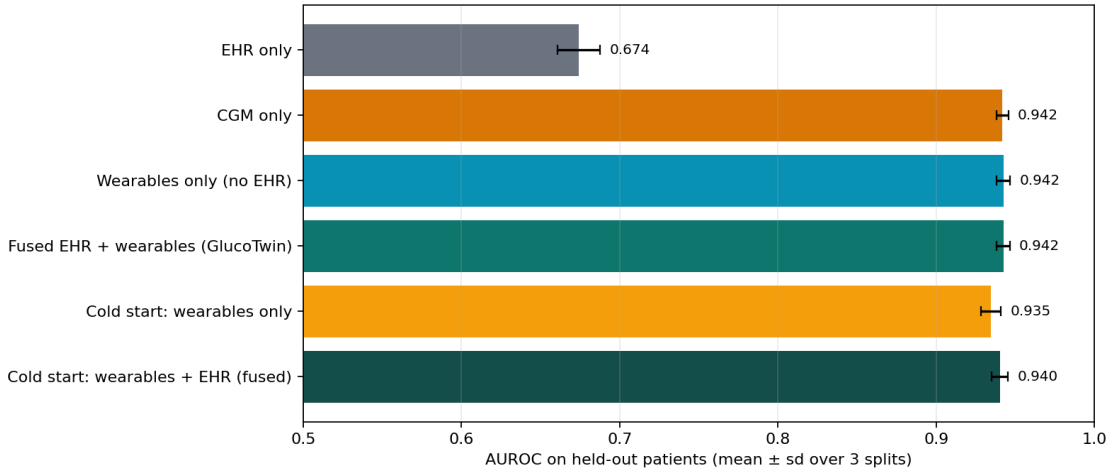
Meal response, insulin resistance, sleep debt, stress and exercise. Re-runs the same meals and noise with one change, so a doctor can ask counter-factual questions.

3

Explain (drivers)

Each forecast shows which input group pushes risk up or down: glucose trend, meal-time habit, sleep, clinical profile, stress, movement.

Results on held-out patients



0.942

AUROC, fused model (3 patient-level splits)

98%

of excursion episodes pre-warned (primary split)

EHR alone: 0.67. Baseline logistic regression on current glucose: 0.61.

Labels come from our simulator, so this measures pipeline learning, not clinical accuracy.

What the ablations actually tell us

1

Glucose history leads

Recent CGM and meal-time habit carry most of the signal (importance 0.18 and 0.11 AUROC). EHR alone cannot time a spike.

2

Fusion helps most at cold start

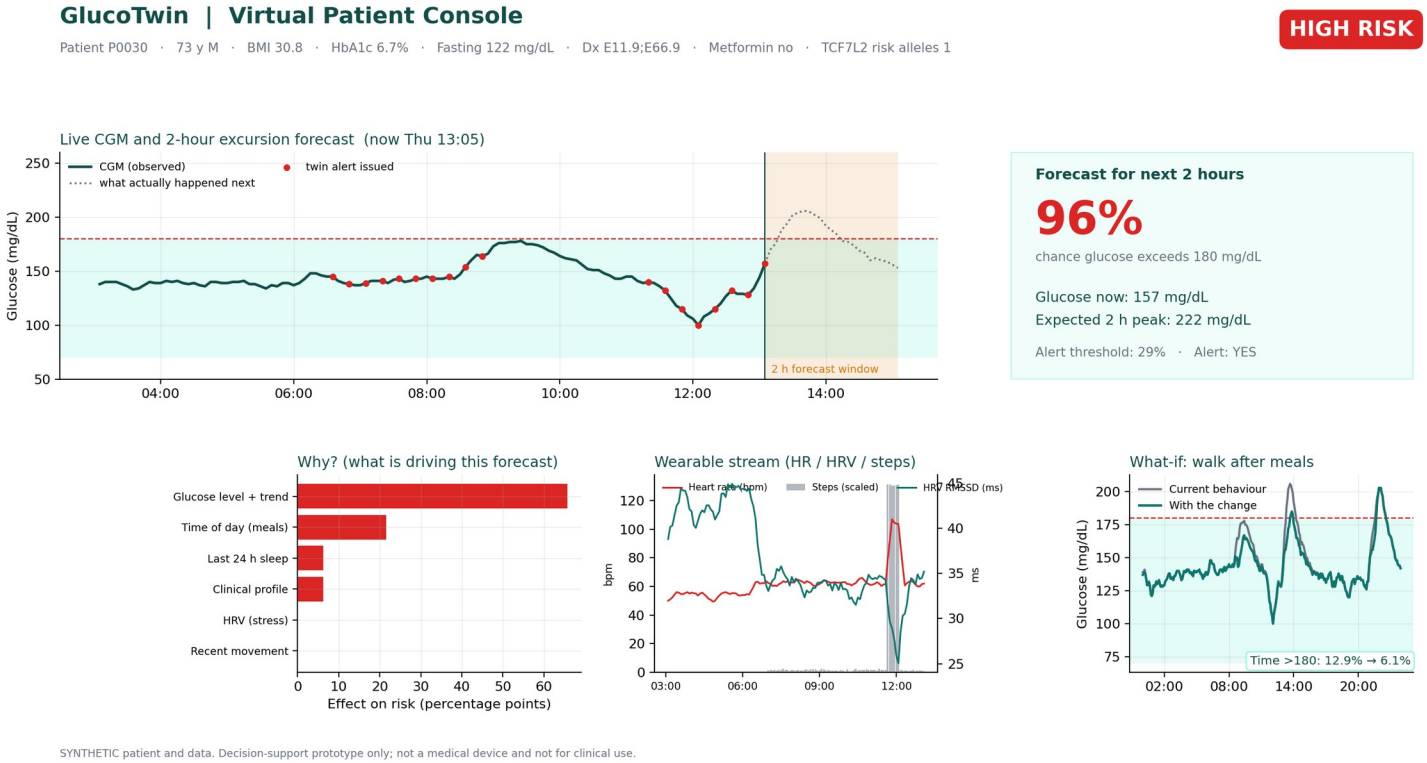
For a new patient with no history, adding the EHR lifts AUPRC from 0.739 to 0.762. With history, the gain is small (0.768 to 0.775).

3

Alerts are too frequent

About 6 per patient-day at 80% sensitivity. Needs clinician-tuned thresholds before any real use.

Conceptual UI: the doctor's console



What-if: walk 15 minutes after meals



12.9% → 6.1%

time above 180 mg/dL (demo patient P0030)

231 → 205

mean daily peak, mg/dL

Same meals, same noise, one change. The effect size reflects simulator assumptions, not clinical evidence.

Limitations and responsible use

- Validated only on synthetic data; metrics are not clinical performance
- Simplified physiology: no insulin dosing, illness, alcohol or sensor drift
- Fixed 180 mg/dL threshold and a high alert rate (alert fatigue risk)
- What-if results are simulations, not prescriptions
- Research prototype, not a medical device, no regulatory clearance

Next steps

- Validate on real open wearable + CGM datasets and anonymised clinical records, with ethics approval
- Estimate each patient's insulin sensitivity from their own data (true personalisation)
- Add uncertainty intervals and clinician-tuned alert policies
- Extend to hypertension and cardiovascular risk using the same twin framework

Thank you · github.com/<your-handle>/TeamName_CollegeName